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Dynamic ultrasound tuning of quantum tunneling barrier height via dielectric dependence

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ABSTRACT

Quantum tunneling has driven considerable advances in theoretical physics, the proliferation of quantum devices, and the development of information technologies. However, the dependence of electron tunneling behavior on the dielectric medium within the tunneling region remains controversial. To this end, a theoretical model for tuning quantum tunneling barrier height through dielectric dependence was established, which comprises the effects of interface dipole, image potential, and external stimuli. Notably, the model revealed a negative exponential relationship between barrier height and dielectric constant. This finding was further experimentally evidenced by using an Au atomic quantum tunneling probe featuring a sub-3 nm gap across different dielectric media. Furthermore, the dynamic control of barrier height was achieved through ultrasound-induced dielectric modulation within the tunneling region. These findings open new avenues for the development of broadband acoustic quantum devices and enable atomic-scale imaging within opaque and biological materials.

I. INTRODUCTION

Electron quantum tunneling,[1] which offers temporal, spatial, and energy resolution, plays a pivotal role in atomic physics analysis,[2] nuclear decay control,[3] semiconductor diode fabrication,[4] superconductor research,[5] and chemical reaction regulation.[6] In metal electrode pairs separated by dielectric films, electron quantum tunneling exhibits an explicit quantitative dependence on characteristic parameters, such as intrinsic field,[7] nanogap,[8] potential barrier (effective height and width),[9] effective tunneling area,[10] and electrode pairs bias voltage.[11,12] This dependence has been extensively studied, forming the basis of the classic tunneling formula.[13,14] However, the relationship between quantum tunneling behavior and tunneling-region dielectric remains controversial. This ambiguity hinders the advancement and application of quantum tunneling devices and thus warrants urgent resolution.[15,16]

One perspective posits that dielectrics with high relative dielectric constant (ε_r) enhance quantum tunneling. Initially, symmetric junctions do not depend on dielectric polarity, in contrast to asymmetric junctions.[7] Defect enrichment at the tunneling junction reduces barrier height, facilitating the transition of dielectric from hard to soft breakdown,[17] and may induce Mott transition.[18] Under tetragonal compression, tunneling junctions (e.g., sub-5 nm-thick epitaxial MgO) exhibit a considerable increase in dielectric constant, which further augments tunneling.[19] Gradient forces that promote flexoelectricity in tunneling dielectric can reduce the tunneling barrier and thereby amplify the tunneling effect.[20] Moreover, a sharp increase in dielectric constant has been shown to enhance quantum tunneling.[21]

An opposing viewpoint contends that high-ε_r dielectrics suppress quantum tunneling. Dual barriers[22] and high tunnel resistance[23] induced by high-ε_r dielectrics would lower tunneling probability. Moreover, strong dielectric screening,[24] considerable dielectric loss,[25] and increased thermal resistance losses[26] inhibit tunneling.

Measurable approaches simultaneously addressing the controversy and resolving the quantitative dependence of tunneling behavior on tunneling-region dielectrics are lacking. To address this gap, we systematically develop a theoretical model for barrier height modulation, which comprises intrinsic field, interface dipole, image potential, and external electric field. We then measure tunneling current–voltage (I–V) characteristics with the same quantum tunneling probe (Au–dielectric–Au with a sub-3 nm gap) with different dielectrics. The results show that the barrier height exhibits a negative exponential dependence on dielectric constant, consistent with theoretical prediction. By leveraging the dynamical modulation of the tunneling-region dielectric through ultrasound-induced stress, we demonstrate ultrasonic detection of subsurface defects in opaque samples. Overall, this study not only resolves long-standing debate surrounding tunneling–dielectric interactions but also establishes a viable method for dynamically modulating electron quantum tunneling. This method can be extended to nanoscale junctions, such as break, stacked, and molecular junctions.

II. METHODS AND RESULTS

A. Dielectric tuning of quantum tunneling barrier height

In a metal–insulator–metal structure, the quantum tunneling barrier height (ø) can be expressed as

$$\o = \o_0 - \o_{ins} - \o_{dipole} - \o_{image} - \o_{bias}, \tag{1}$$

where $\o_0$, $\o_{ins}$, $\o_{dipole}$, $\o_{image}$, and $\o_{bias}$ are the initial barrier height, electrode's intrinsic properties, interface dipoles, image potential, and external field modulation, respectively. The initial barrier height $\o_0$ for electrons transitioning from the metal Fermi level to the dielectric's conduction band minimum is governed by their energy offset, which is expressed as follows:[27,28]

$$\o_0 = \psi - \chi, \tag{2}$$

where ψ is the work function of metal electrodes, and χ is the electron affinity of dielectrics.

Driven by the difference in work function between the two metal electrodes, electrons spontaneously tunnel from the electrode with lower work function (ψ_1) to a higher one (ψ_2). Accordingly, an intrinsic field (i.e., the contact potential difference) is established at the equilibrium, which can be described by the following equation:[7]

$$\o_{ins} = \frac{\psi_2 - \psi_1}{d}, \tag{3}$$

where d denotes the width of nanogap and is approximately equal to the barrier width. The intrinsic, localized interface dipole arises from work function difference, Fermi-level alignment, and interface states at the metal–dielectric interface, leading to charge transfer and electron cloud overlap and forming an ultrathin charge separation layer that can be quantified,[29,30]

$$\rho_{eff} = \frac{\varepsilon_r - 1}{\varepsilon_r}\rho_0, \tag{4}$$

$$p_{int} = \rho_{eff}\delta, \tag{5}$$

$$V_{dipole} = \frac{p_{int}}{\varepsilon_0 \varepsilon_r}, \tag{6}$$

where ρ_{eff} is the effective interface charge density, ρ_0 is the vacuum interface charge density, p_{int} is the interface dipole area density, δ is the dipole length, and V_{dipole} is the interface dipole potential.

Substituting Eqs. (4) and (5) into Eq. (6), we obtain $V_{dipole} = \frac{\rho_0\delta}{\varepsilon_0}\left(\frac{1}{\varepsilon_r} - \frac{1}{\varepsilon_r^2}\right)$. For further simplification, we apply $e^x = \sum_{n=0}^{\infty}\frac{1}{n!}x^n, x \in (-\infty, +\infty)$ with $x = -\frac{1}{\varepsilon_r}$, and retain only the dominant terms. Equation (6) can thus be approximated as

$$V_{dipole} \approx \frac{\rho_0\delta}{\varepsilon_0}\left(1 - e^{-\frac{1}{\varepsilon_r}}\right). \tag{7}$$

Furthermore, a size factor F_{scale} is defined to account for the size effect, which gives

$$F_{scale} = \frac{\delta}{d}[1 + \mu(\varepsilon_r - 1)], \tag{8}$$

where F_{scale} quantifies the synergistic amplification of the interface dipole effect by geometric confinement ($\frac{\delta}{d}$) and dielectric response $[\mu(\varepsilon_r - 1)]$. In traditional systems, the interface dipole is diluted within the bulk medium, and the bulk polarization is larger ($\delta \ll d$). However, in quantum tunneling architectures ($\delta \to d$), the interface effect constitutes a considerable portion of the dielectric layer. In this context, it cannot be adequately approximated by bulk properties but directly governs the overall potential field and electrical characteristics. Furthermore, the dimensionless factor μ (typically ranging from 1.0 to 1.3) quantifies the enhancement of the relative permittivity ε_r because of nanoscale effects.[19] Specifically, it originates from a feedback mechanism: The electric field generated by the interface dipole induces dielectric polarization, and the resulting polarization field in turn reinforces the original dipole moment. Accordingly, μ denotes the linear response of the interface dipole moment to this self-consistent dielectric polarization at room temperature.

Hence, the interface dipole barrier evolves as

$$\o_{dipole} = eV_{dipole}F_{scale}. \tag{9}$$

Meanwhile, the image potential ($\o_{image}$)[31] and external electric field ($\o_{bias}$)[32] can be determined using the following equations:

$$\o_{image} = \sqrt{\frac{e^3 V_{bias}}{4\pi\varepsilon_0\varepsilon_r d}}, \tag{10}$$

$$\o_{bias} = eV_{bias}. \tag{11}$$

After Eqs. (2), (3), and (9)–(11) were substituted into Eq. (1), the contribution from Eq. (2) becomes constant, and the contact potential difference ($\o_{ins}$) vanishes, that is, $\o_{ins} = 0$, for the symmetric electrode pair with a dielectric layer (identical in geometry, atomic configuration, and material orientation). Furthermore, the linear potential field induced by an external bias is treated under the low-bias approximation.[31] Strong coupling occurs between interface dipole potentials when $d \le 2\delta$, and the image potential ($\o_{image}$) diverges at the boundaries. As a result, $\o_{image}$ is suppressed in dielectrics with a high ε_r, whereas the interface dipole potential becomes dominant. Consequently, we derived the simplified relationship as $\o \propto -e^{-\frac{1}{\varepsilon_r}}$. In this sense, the quantum tunneling current density can be evaluated by applying WKB approximation to the Schrödinger equation with a hyperbolic barrier.[31]

To validate the theoretical model for tuning barrier height through dielectric dependence, a quantum tunneling probe was fabricated using a tapered tip derived from the double-barrel theta quartz capillaries as the substrate. Figures 1(a) and 1(b) show the overall structure and tip of the probe, respectively. The fabrication process proceeded sequentially following butane cracking carbon deposition, carbon etching, feedback deposition of Au atoms, and surface reformation of Au clusters with de-ionized water. This process yielded a probe with a sub-3 nm gap, where the diameter of the Au electrodes was ∼100 nm (supplementary material S1).[10] In this Au–dielectric–Au electrode structure, the top of the dielectric's bandgap lies above the Fermi level of the Au electrodes. Thus, the dielectric layer introduces a potential barrier that inhibits electron transport. However, in a sufficiently thin intermediate dielectric layer, electrons can tunnel through its potential barrier from one Au electrode to another. This behavior is driven by the intrinsic properties of the junction and external modulation. The net tunneling current corresponds to the net flow of electrons, which is defined as the difference between the forward (left-to-right) and backward (right-to-left) electron tunneling rates across the barrier, as illustrated in Fig. 1(b). Meanwhile, to ensure consistency with experimental measurements, we visualized the hyperbolic barrier setting model [Fig. 1(c)].

Using the sub-3 nm quantum tunneling probe, the current–voltage (CV) curve was first measured with air gap (Au–air–Au) by sweeping DC bias from −300 to 300 mV 200 times, as shown in Fig. 2(a) (blue solid lines). Notably, the potential barrier retained a quasi-rectangular shape under low biases ranging from −100 to 100 mV, which led to an approximately linear tunneling current. As the bias increased, the barrier shape was initially trapezoidal ($100\,\mathrm{mV} \le |V| \le 200\,\mathrm{mV}$) and then became triangular ($|V| > 200\,\mathrm{mV}$), and a pronounced nonlinearity emerged in the CV curve. The bias was truncated at 300 mV because it may increase the risk of dielectric breakdown, which may permanently damage the tunnel junction. These attributes are consistent with previous studies on nanoscale tunneling junctions.[7,31,48–50] For quantitative analysis, Simmons' model was used to yield the statistical distribution of barrier height with 200 CV measurements. The measurements were repeated with multiple probes fabricated from the same patch. Figure 2(b) shows the result of barrier height with air. Detailed procedures for fitting and data processing procedures based on Simmons' model are provided in S3 of the supplementary material. The gap width was smaller than 3 nm, consistent with the sub-3 nm quantum tunneling probe. Similarly, the barrier height distributions for the other dielectrics are presented in S4 of the supplementary material.

To characterize the effect of dielectric constant (ε_r) on tunneling, we filled the same quantum tunneling probe with different dielectrics. The dielectric constant ranged from 1.0 to 101, and the other factors were unchanged. The average CV curves obtained with different dielectrics are illustrated in Fig. 2(c). The tunneling current under the same bias increased monotonically with the dielectric constant, providing direct evidence that the tunneling enhancement was dielectric dependent. The average barrier height of different dielectrics was then obtained. Figure 2(d) reveals a negatively exponential relationship between barrier height and dielectric constant, which fits ε_r and barrier height with $a\varepsilon_r^b$, where a and b are the fitting constants: $a = 1.05 \pm 0.12$ and $b = -0.25 \pm 0.05$. The dielectric constants used throughout this study are under the static (low frequency) assumption, consistent with the DC bias applied across the tunneling junction.

B. Dynamic ultrasound tuning of quantum tunneling barrier height

Given the effect of pressure on the dielectric constant,[19,20] dynamically tuning barrier height through ultrasound-induced

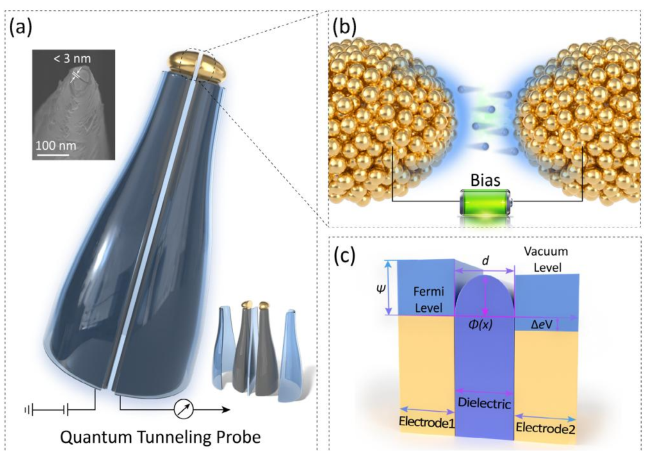

FIG. 1. Schematic of the quantum tunneling probe structure and electron tunneling across the barrier. (a) Schematic of quantum tunneling probe with a sub-3 nm gap [upper left: scanning electron microscopy (SEM) image of the quantum tunneling probe tip; lower right: overall diagram of the probe structure]; (b) diagram of electron energy distribution and tunneling in the Au–dielectric–Au quantum probe under bias voltage; and (c) schematic of the quantum tunneling barrier in a metal–dielectric–metal (MDM) structure with external modulation.

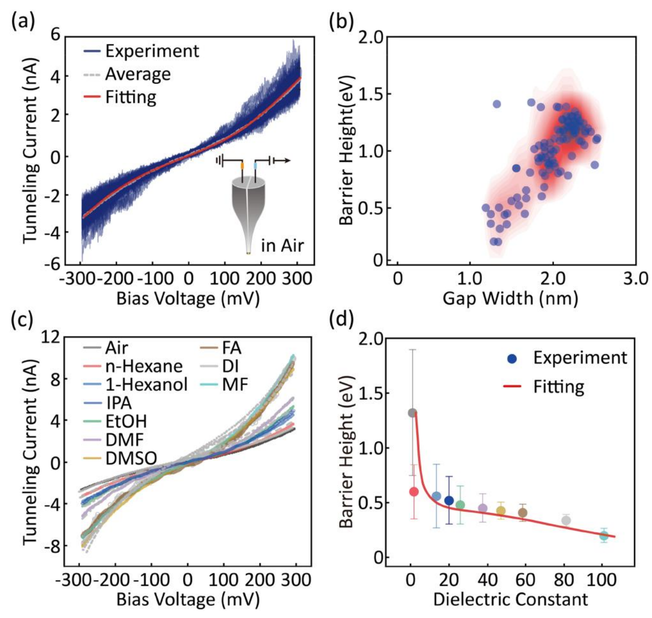

FIG. 2. Dielectric dependence measurement and fitting for quantum tunneling probe with a sub-3 nm gap. (a) Measurement of the tunneling current vs bias voltage (CV curves) for the quantum tunneling probe in air, and a superposition of 200 experimental traces (blue) is shown, with the average (gray, dotted) and its fitting (red, solid); (b) the distribution of barrier height on gap width obtained by fitting every CV curve (total 200 measured curves) in the air dielectric [from panel (a)]; (c) the plots of the averaged CV curves for the same quantum tunneling probe in different dielectrics; and (d) dependence of barrier height on dielectric constant. Experimental points represent air,[33] n-Hexane,[34] 1-Hexanol,[35,36] IPA,[37,38] EtOH,[39,40] DMF,[35,36,40,41] DMSO,[40,42] FA,[43] DI,[40,44] and MF,[45–47] respectively (details in the supplementary material S2).

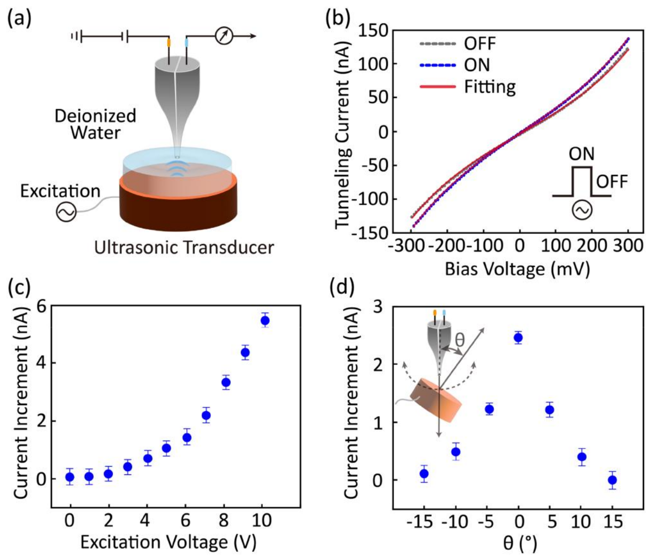

FIG. 3. Measurement of quantum tunneling current under ultrasound modulation. (a) Schematic of ultrasound modulation for quantum tunneling in de-ionized water; (b) comparison of CV curves for quantum tunneling probe with ultrasonic excitation on (blue, dotted), off (gray, dotted), and its fitting (red, solid); (c) the demodulated tunneling current signal as a function of stepwise-increased ultrasonic excitation intensity (frequency $f = 1$ MHz), measured at a bias voltage of 100 mV; and (d) directional dependence of ultrasonic longitudinal waves incident on quantum tunneling probe (direction defined as the acute angle between the normal to the transducer excitation surface and the longitudinal central axis of the probe).

mechanical vibrations is feasible. The quantum tunneling probe was immersed in de-ionized water for CV curve measurements under ultrasound excitation ($f = 1$ MHz). As shown in Fig. 3(b), the tunneling current increased but at a lower rate than the increase observed in the absence of ultrasound excitation. To quantify this effect, we gradually increased the acoustic intensity by incrementally adjusting the excitation voltage of ultrasonic transducer from 1 to 10 V, and the tunneling current was recorded under a fixed bias of 100 mV. Figure 3(c) shows that the tunneling current increment increases with the stepwise increments of acoustic intensity, indicating synchronous coupling between macroscopic ultrasound vibrations and nanoscale tunneling. Figure 3(d) illustrates this coupling at varying angles of acoustic incidence (θ). At the same tip-transducer distance, the tunneling current progressively decreased as θ changes from 0° (normal incidence) to 15° (oblique incidence). This change was due to the angular dependence of acoustic intensity. Moreover, as acoustic pressure increased (σ_0), the amplitude of the periodic strain exhibited a dependence, which led to an approximately proportional variation in the dielectric constant ($\Delta\varepsilon_r$). Under weak modulation, the corresponding decrease in the tunneling barrier height (ø) was nearly linear, resulting in an enhanced tunneling current density (ΔJ). These results are consistent with the theoretical model, which is detailed in the Appendix.

To primarily demonstrate the application with ultrasound-induced dynamic tuning of quantum tunneling barrier height, we carried out a proof-of-concept experiment for detecting subsurface defect in an aluminum sample. The experimental setup is schematically illustrated in Fig. 4(a). Herein, the subsurface defects were artificially

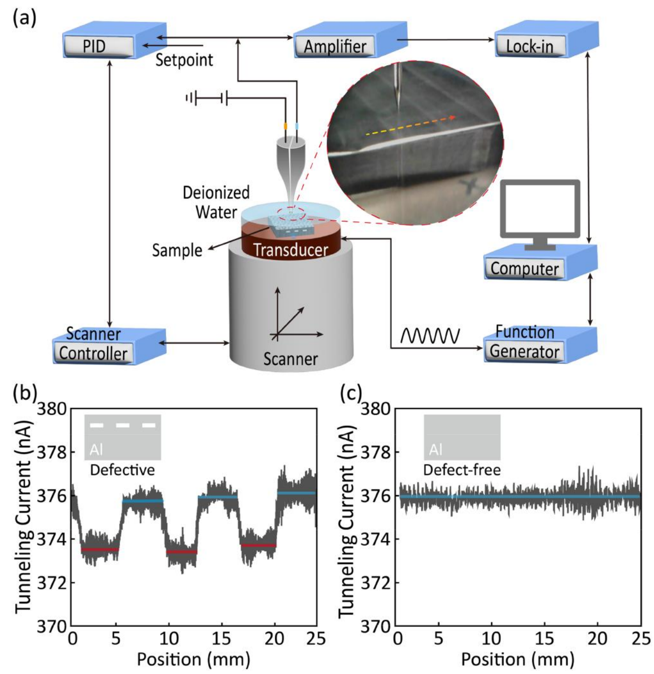

FIG. 4. Detection of subsurface defects using a quantum tunneling probe. (a) Schematic of the detection system for embedded subsurface defects (burial depth: 2 mm) in an aluminum sample; (b) comparison of detection signals with (red line) and without (blue line) a subsurface defect; and (c) detection result for a uniform, defect-free sample (control experiment).

prepared by burying rectangular cavities at a depth of 2 mm. As shown in Fig. 4(b), the signals from the defective regions exhibited a distinct reduction in tunneling current, whereas the signals obtained from the reference sample without defects remained nearly unchanged [Fig. 4(c)]. This difference was expected because fluctuations in tunneling current originated from acoustic scatterings at the defective regions, which perturbed the dielectric polarization in the tunneling gap that modulated the barrier height. This effect enables the identification of subsurface defects. The flow chart of signal processing is provided in S5 of the supplementary material.

III. DISCUSSION

The theoretical and experimental results presented above resolve the long-standing ambiguity regarding dielectric–tunneling interactions and yield three definitive conclusions. First, quantum tunneling exhibits a strong, quantifiable dependence on the tunneling-region dielectric, consistent with existing studies.[7,21] Second, tunneling barrier height decreases exponentially with the dielectric constant. Finally, the tunneling barrier height can be modulated by ultrasound-induced mechanical stress, macroscopically enabling the dynamic control of quantum tunneling.

As the net tunneling current arises from the competition between electron escape, tunneling, and bidirectional capture within the electrode junction,[51,52] the measurement environment, gap width, and effective area of probe were not changed. Barrier height is the dominant factor governing the tunneling characteristics[26,53] owing to metal work function, dielectric electron affinity, interface dipole, image potential, and external modulation.

The quantum tunneling probe was fabricated through chronopotentiometry with controlled electro-deposition of Au atoms and steady-state growth,[10] yielding a symmetrical geometric structure that suppresses energy dissipation.[54–56] This symmetry ensures barrier symmetry and stability. After a symmetric bias of ± 300 mV was applied to the probe, the tunneling current exhibited approximate polarity symmetry without discernible directional dependence [Fig. 2(a)]. This finding confirms that junction symmetry does not contribute to barrier height variations.

In the sub-3 nm gap, the dielectrics of different constants modulate the Au–dielectric–Au interface dipole, which in turn tunes the electron tunneling barrier height. Specifically, high ε_r enhances the interface dipole potential and therefore increases the electron tunneling current. This increase is amplified by F_{scale}. Notably, F_{scale} quantifies the nanoscale confinement effect when the gap width decreases below 3 nm, which would further enhance interface dipole potential to lower barrier height. This effect explains why dependence between barrier height and dielectric constant was only observed in sub-3 nm junctions [Fig. 2(d)]. Therefore, nanoscale gaps are a prerequisite for interface dipole-dominated barrier modulation.

The ultrasound-induced tunneling current enhancement is due to the transient, spatiotemporal modulation of barrier height by mechanical stress. Different from molecular junctions that rely on bulk dielectric barrier tuning, the quantum sensing system with a sub-3 nm gap responds to ultrasound-induced pressure,[57] and tunneling current enhancement scales linearly with acoustic intensity. This phenomenon is further confirmed at varying acoustic intensities.

In addition, the proposed framework of tunneling-region barrier height modulation via dielectric properties can be generalized to broad scenarios. At the nanoscopic scale, the dielectric polarity,[58] interface state density,[59,60] and thermal expansion can tune the barrier height in heterogeneous junctions (e.g., metal–oxide–semiconductor[55]). At the molecular scale, single molecular chain length,[21,61] conformational isomers,[62] and short- or long-range intermolecular interactions[20] can modify barrier width or height.[63] These extensions leverage the core finding, namely, dielectric-dependent barrier height modulation, and therefore would advance quantum device design and atomic-scale characterization.

Given the frequency dependence of the dielectric response, a high-frequency bias excitation may perturb the tunneling barrier height or width to introduce noise, arising from the interfacial electron energy distribution, local Joule heating, and time-dependent polarization relaxation. Subsequently, electron tunneling and ultrasonic modulation would be influenced.

IV. CONCLUSIONS

A theoretical model for quantum tunneling barrier height was developed; it considered the probe's intrinsic properties, interface dipole, image potential, and external electric field. It resolved the long-standing debate surrounding the relationship between barrier height and dielectric constant. Using the quantum tunneling probe with a sub-3 nm gap, we investigated the negative exponential dependence of barrier height on the dielectric constant, and the results were consistent with the model predictions. Moreover, barrier height modulation by high dielectric was primarily attributed to the formation of an interface dipole-induced depolarization, which effectively reduced the barrier height and enhanced quantum tunneling. Meanwhile, a novel approach for the dynamic control of barrier height by ultrasound was proposed. Leveraging the dielectric modulation of the tunneling region to regulate the barrier height, this strategy is applicable to quantum sensing, quantum metrology, and quantum information processing[64,65] and may contribute to the development of highly sensitive ultrasonic quantum sensors[66] and nondestructive characterization of subsurface atomic features.[67]

SUPPLEMENTARY MATERIAL

See the supplementary material for S_Fig.1–S_Fig.3 and S_Table1.

ACKNOWLEDGMENTS

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ye Tian: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). **Biao-Feng Zeng:** Formal analysis (supporting); Investigation (supporting); Methodology (supporting). **Mengru Zhang:** Investigation (supporting); Supervision (supporting). **Long Yi:** Investigation (supporting); Supervision (supporting). **Longhua Tang:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Jian Chen:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

APPENDIX: ELECTRON QUANTUM TUNNELING WITH ULTRASONIC DIELECTRIC MODULATION

1. Ultrasound modulation on dielectric permittivity

The periodic stress field $\sigma(t)$ generated by the ultrasound wave can be represented by

$$\sigma(t) = \sigma_0 \cos(\omega t),$$

where σ_0 and ω are the amplitude and angular frequency of the ultrasound wave.

According to the electroelastic theory,[68,69] the dielectric constant varies as

$$\Delta\varepsilon_r(t) = \gamma\sigma(t),$$

where $\gamma = \left(\frac{\partial\varepsilon_r}{\partial\sigma}\right)$ is the electroelastic coefficient that governs the efficiency of stress-induced modulation on the dielectric constant.

Therefore, the effective dielectric constant ε_r^{eff} can be derived as

$$\varepsilon_r^{eff} = \varepsilon_{r0} + \Delta\varepsilon_r(t) = \varepsilon_{r0} + \gamma\sigma_0\cos(\omega t),$$

where ε_{r0} is the relative permittivity of the dielectric.

2. The relationship between dielectric constant and electric field

Under a constant bias voltage (V_{bias}), the initial electric field (E_0) and electric displacement field (D_0) in the quantum tunneling probe are

$$E_0 = \frac{V_{bias}}{d},$$
$$D_0 = \varepsilon_0\varepsilon_{r0}E_0,$$

where d is the gap width between tunneling electrodes.

The transient electric displacement field (D) in the quantum tunneling probe is

$$D = \varepsilon_0\varepsilon_r^{eff}(t)E(t).$$

When the $\varepsilon_r(t)$ dynamically changes while the electric displacement field remains constant ($D = D_0$), the field strength $E(t)$ is

$$E(t) = \frac{\varepsilon_{r0}E_0}{\varepsilon_{r0} + \gamma\sigma_0\cos(\omega t)}.$$

With small perturbation ($\gamma\sigma_0 \ll \varepsilon_{r0}$), $E(t)$ can be approximated as

$$E(t) \approx E_0\left[1 - \frac{\gamma\sigma_0}{\varepsilon_{r0}}\cos(\omega t)\right].$$

Then the periodic induced electric field variation $\Delta E(t)$ in the dielectric is

$$\Delta E(t) \approx -\frac{\gamma}{\varepsilon_{r0}}\sigma_0\cos(\omega t).$$

3. The mutagenic electric field modulation on tunneling barrier height

The effective barrier height $\o(t)$ in the quantum tunneling probe is

$$\o(t) = \o_0 - \Delta\o(t),$$

where $\o_0$ is the initial static barrier height, $\Delta\o(t)$ is the reduction of barrier height.

Based on the image potential in metal–insulator–metal structure,[31,70] the $\Delta\o(t)$ can be defined as

$$\Delta\o(t) = \sqrt{\frac{e^3E(t)}{4\pi\varepsilon_0\varepsilon_{r0}}}.$$

As $\Delta E(t) \ll E_0$, the periodic modulation of barrier height follows:

$$\Delta\o(t) \approx \frac{\partial[\Delta\o(t)]}{\partial E}\bigg|_{E=E_0}\Delta E(t) = -\eta\sigma_0\cos(\omega t),$$

where $\eta = \frac{\gamma}{2\varepsilon_{r0}}\sqrt{\frac{e^3E_0}{4\pi\varepsilon_0\varepsilon_{r0}}}$ is the comprehensive coefficient, which represents the modulation efficiency of the tunnel barrier height by periodic stress.

4. The tunneling current density $J(t)$ under barrier height modulation

A generalized formalism for tunneling current density in metal–insulator–metal structures can be written as[10,31]

$$J(t) \propto E_0^2\exp\left(-\frac{4\sqrt{2m^*}[\o(t)]^{3/2}}{3e\hbar E_0}\right),$$

where m^* denotes the effective electron mass, $\hbar = \frac{h}{2\pi}$, with h being Planck's constant.

Owing to the dominant role of the exponential term, the variation of current density $\Delta J(t)$ can be derived as

$$\frac{\Delta J(t)}{J_0} \approx \frac{\partial[\ln J(t)]}{\partial\o(t)}\bigg|_{\o(t)=\o_0}\Delta\o(t),$$

where J_0 is the static tunneling current density.

Taking the logarithmic derivative, it yields

$$\frac{\partial[\ln J(t)]}{\partial\o(t)} = -\frac{2\sqrt{2m^*\o(t)}}{e\hbar E_0}.$$

Therefore, we can obtain

$$\frac{\Delta J(t)}{J_0} \approx \xi\Delta\o(t),$$

where $\xi = \frac{2\sqrt{2m^*\o_0}}{e\hbar E_0}$.

The ultrasonic quantum sensing signals are then integrated as

$$\frac{\Delta J(t)}{J_0} \approx -\xi\eta\sigma_0\cos(\omega t).$$

That is,

$$\Delta J(t) \propto J_0\sigma_0\cos(\omega t).$$

As a result, one can realize high-sensitivity ultrasonic detection, i.e., ultrasonic quantum sensing, by measuring the tunneling current synchronized to the ultrasonic frequency under slight perturbation.

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